

How to Use groupby() and transform() Functions in Pandas

BY ZACH BOBBITT • MAY 27, 2022

You can use the following methods to use the **groupby()** and **transform()** functions together in a pandas DataFrame:

Method 1: Use groupby() and transform() with built-in function

```
df['new'] = df.groupby('group_var')['value_var'].transform('mean')
```

Method 2: Use groupby() and transform() with custom function

```
df['new'] = df.groupby('group_var')['value_var'].transform(lambda x: some_function)
```

The following examples show how to use each method in practice with the following pandas DataFrame:

```
import pandas as pd

#create DataFrame
df = pd.DataFrame({'team': ['A', 'A', 'A', 'A', 'B', 'B', 'B', 'B'],
                  'points': [30, 22, 19, 14, 14, 11, 20, 28]})

#view DataFrame
print(df)

   team  points
0    A     30
1    A     22
2    A     19
3    A     14
4    B     14
5    B     11
6    B     20
7    B     28
```

Example 1: Use groupby() and transform() with built-in function

The following code shows how to use the **groupby()** and **transform()** functions to add a new column to the DataFrame called mean_points:

```
#create new column called mean_points
df['mean_points'] = df.groupby('team')['points'].transform('mean')

#view updated DataFrame
print(df)

   team  points  mean_points
0    A     30         21.25
1    A     22         21.25
2    A     19         21.25
3    A     14         21.25
4    B     14         18.25
5    B     11         18.25
6    B     20         18.25
7    B     28         18.25
```

The mean points value for players on team A was **21.25** and the mean points value for players on team B was **18.25**, so these values were assigned accordingly to each player in a new column.

Note that we could also use another built-in function such as **sum()** to create a new column that shows the sum of points scored for each team:

```
#create new column called sum_points
df['sum_points'] = df.groupby('team')['points'].transform('sum')

#view updated DataFrame
print(df)

   team  points  sum_points
0    A     30           85
1    A     22           85
2    A     19           85
3    A     14           85
4    B     14           73
5    B     11           73
6    B     20           73
7    B     28           73
```

The sum of points for players on team A was **85** and the sum of points for players on team B was **73**, so these values were assigned accordingly to each player in a new column.

Example 2: Use groupby() and transform() with custom function

The following code shows how to use the **groupby()** and **transform()** functions to create a custom function that calculates the percentage of total points scored by each player on their respective teams:

```
#create new column called percent_of_points
df['percent_of_points'] = df.groupby('team')['points'].transform(lambda x: x/x.sum())

#view updated DataFrame
print(df)

   team  points  percent_of_points
0    A     30         0.352941
1    A     22         0.258824
2    A     19         0.223529
3    A     14         0.164706
4    B     14         0.191781
5    B     11         0.150685
6    B     20         0.273973
7    B     28         0.383562
```

Here's how to interpret the output:

- The first player on team A scored 30 out of 85 total points among team A players. Thus, his percentage of total points scored was $30/85 =$ **0.352941**.
- The second player on team A scored 22 out of 85 total points among team A players. Thus, his percentage of total points scored was $22/85 =$ **0.258824**.

And so on.

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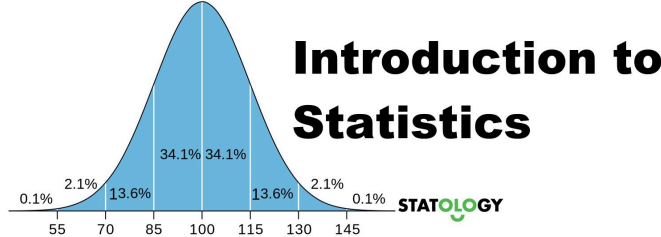
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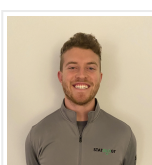
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Note that we can use the **lambda** argument within the **transform()** function to perform any custom calculation that we'd like.

Additional Resources

The following tutorials explain how to perform other common operations in **pandas**:

- [How to Perform a GroupBy Sum in Pandas](#)
- [How to Use Groupby and Plot in Pandas](#)
- [How to Count Unique Values Using GroupBy in Pandas](#)



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